

Research Methodology

Research Project Description

Introduction

Welcome to your final project!

Students must complete the project step by step.

- ✓ The first required step is to prepare a research design proposal for review.
- ✓ After the proposal is reviewed, students will conduct the research, analyze the findings, write the research paper, and prepare presentation slides

The project must be completed in groups.

Good luck !

Agenda & Deliverables

DATE	EVENT	DELIVERABLES
20 June 2026	Research Topic Proposal Review	Research design proposal for review on 21 June 2026
5 July 2026	Paper and Slide Material Submission	Final research paper and presentation slides
(To be decided)	Presentation and Paper Resubmission	Research presentation and revised paper, if required

1. Project Overview

Each group must select a suitable research topic related to your specialization in **Artificial Intelligence (AI) and Data Science**. The topic should focus on a clear issue, application, trend, method, or problem in areas such as machine learning, data analytics, data visualization, natural language processing, computer vision, predictive modelling, recommender systems, big data, or ethical issues in AI and data science.

Each group must design a clear research plan, conduct a short research study, and submit the required academic materials. The final research paper may be written as either:

- **a review paper**, which analyses and compares existing studies, methods, technologies, datasets, or applications related to a selected AI or Data Science topic; or
- **an original research paper**, which presents a small-scale study involving data collection, data analysis, experiment design, model testing, survey results, or case study findings.

The research project should demonstrate the students' ability to:

- identify a clear, relevant, and researchable topic in AI or Data Science;
- formulate a problem statement, research objectives, and research questions;
- design an appropriate research methodology based on the selected paper type;
- collect and analyse relevant data, evidence, literature, or experimental results;
- write a structured academic research paper using an appropriate academic style;
- present the research process, key findings, and conclusions clearly.

2. Research Design Requirement

Students must prepare the research design before starting the full research. The research design acts as the project proposal and must be reviewed on 21 June 2025.

The research design should include the following components:

- research title;
- background of the study;
- problem statement;
- research objectives;
- research questions;
- significance and scope of the study;
- research methodology and data analysis plan;
- expected findings;
- research timeline;
- references.

Students should not begin the full research process until the research design has been reviewed. The lecturer may provide comments and request revisions before students continue.

3. Research Project Process

1. Choose a research topic related to the course or an approved area of study.
2. Prepare the research design proposal using the provided proposal template.
3. Submit and discuss the proposal during the review session on 21 June 2025.
4. Revise the proposal if feedback is provided by the lecturer.
5. Conduct the research over approximately two weeks after the proposal review.
6. Analyze the collected data or evidence.
7. Write the final research paper and prepare presentation slides.
- Present the research project and resubmit the paper if required.

4. Final Research Paper Requirements

The final research paper must be based on the approved research design. The paper should be organized, evidence-based, and written in a clear academic style.

The recommended paper structure is:

- Title and author information;
- Abstract;
- Introduction;
- Literature review or related studies;
- Research methodology;
- Findings and analysis;
- Discussion;
- Conclusion and recommendations;
- References;
- Appendices, if needed.

5. Presentation Slide Requirements

The research design should include the following components: The presentation slides should summarize the research project in a clear and concise way. Slides should focus on the research topic, methodology, key findings, and conclusion.

The recommended slide content includes:

- research title and team members;
- background and problem statement;
- research objectives and questions;
- methodology;
- key findings;
- conclusion and recommendations;
- references or acknowledgement, if needed.

6. Project Evaluation Focus

A/ Research Design and Methodology
The research topic is clear, realistic, and researchable.
The problem statement, objectives, and research questions are logically connected.
The methodology is suitable for answering the research questions.
The research design is completed before the research work begins.
B/ Research Paper Quality
The paper follows a clear academic structure.
The analysis is relevant to the research objectives and questions.
The writing is clear, organized, and properly referenced.
C/ Presentation
The presentation explains the research process and findings clearly.
The slides are well organized and support the oral presentation.
Students can answer questions about their research work.
D/ Student Involvement
Each student contributes to the research project.
For group work, tasks are divided fairly and each member can explain their contribution.

Final Reminder

Students should choose a topic that can realistically be completed within the given timeline. A strong research design will make the two-week research and writing period more manageable and effective.